DriveStateReg: A Time Window-Level State Recognition Framework for Driving Behavior Analysis

Xinyuan Zhao
Shaojie Min
College of Computer Science and Artificial Intelligence,
Fudan University
Shanghai, China
xinyuanzhao25@m.fudan.edu.cn
sjmin24@m.fudan.edu.cn

Wenping Duan
Yang Zhang
Qin Su
Seres Auto Co., Ltd.
Chongqing and Shanghai, China
wenping.duan@seres.cn
yang.zhang215996@seres.cn
qin.su718763@seres.cn

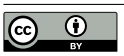
Qiang Duan
Department of Information Sciences and Technology,
Pennsylvania State University
Abington, PA, USA
qduan@psu.edu

Xin Wang
Yang Chen*
College of Computer Science and Artificial Intelligence,
Fudan University
Shanghai, China
xinw@fudan.edu.cn
chenyang@fudan.edu.cn

Abstract

Driving behavior analysis serves as a critical foundation for modern intelligent transportation, enabling vital applications such as road safety enhancement and personalized driver-assistance systems. While prior research frequently analyzes driving behavior through short time windows, widely used datasets are commonly annotated only at the trip level. This creates a granularity mismatch: a trip may contain diverse driving states that cannot be faithfully represented by a single trip-level category. Despite this mismatch, existing methods often directly inherit trip-level labels for time-window analysis. Such direct inheritance obscures behavioral heterogeneity and yields substantial feature space overlap, providing weak and conflicting supervision for time window behavior modeling. In this work, we empirically examine this label-granularity mismatch. To address this problem, we propose DriveStateReg, a state recognition framework that infers behavioral states from unlabeled ego-vehicle motion signals through unsupervised clustering, LLM-based annotation, and manual verification. Applying this framework to a real-world driving dataset successfully identifies fine-grained driving state categories that better characterize time window-level driving behaviors and reveal more interpretable within-trip transition patterns. Compared with inherited trip labels, the states identified by DriveStateReg increase the Silhouette score from −0.01 to 0.14, reduce the Davies-Bouldin score from 9.94 to 2.10, and increase KNN purity@10 from 0.54 to 0.89.

*Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License.
UbiComp Companion '26, Shanghai, China
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2533-3/2026/10
https://doi.org/10.1145/3798063.3837214

CCS Concepts

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**.

Keywords

Driving Behavior, Unsupervised Learning, State Recognition, Large Language Models, Semantic Annotation

ACM Reference Format:
Xinyuan Zhao, Shaojie Min, Wenping Duan, Yang Zhang, Qin Su, Qiang Duan, Xin Wang, and Yang Chen. 2026. DriveStateReg: A Time Window-Level State Recognition Framework for Driving Behavior Analysis. In *Companion of the 2026 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '26), October 11–15, 2026, Shanghai, China.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3798063.3837214

1 Introduction

Driving behavior characterizes the operational patterns a driver exhibits when controlling a vehicle and responding to traffic situations. Understanding such patterns is important for road safety [3, 16, 28], intelligent transportation [4, 8, 34], and behavior-aware driver-assistance systems [33]. A key issue in driving behavior analysis is temporal granularity. Long-term driving behavior describes a driver's habitual tendency across multiple trips or road contexts, and is often used to explain risk perception and individual differences [28]. However, a long-term tendency does not imply that every short time window within a trip exhibits the same behavior. Existing public vehicle-sensing datasets for driving behavior analysis, however, often provide labels only for entire trips. For example, UAH-DriveSet [25], a representative public vehicle-sensing dataset widely used in driving behavior analysis, provides trip-level driving categories, namely Aggressive, Normal, and Drowsy. These labels describe the driving condition of the entire trip but lack manual time window-level annotations. Nevertheless, many existing

analysis methods segment sensor streams into short windows and inherit the label from the trip [11, 19, 21, 29]. In real-world driving, however, the driving process is usually not a continuous repetition of a single behavior. Thus, direct inheritance of trip-level labels may provide weak and conflicting supervision for time window-level behavior modeling. Section 2 empirically examines this problem. In this paper, we use the term *inherited trip labels* to refer to the baseline obtained by directly inheriting trip-level labels to time windows.

To address this problem, we propose DriveStateReg, a framework that discovers behavioral states from unlabeled ego-vehicle dynamics, and assigns semantic interpretations through LLM-based annotation and manual verification. Applying DriveStateReg to UAH-DriveSet, we identify fine-grained driving states that provide a more refined categorization of time window-level driving behaviors than the inherited trip labels, and reveal how the distribution and transitions of these states characterize within-trip behavioral patterns. The main contributions of this paper are as follows.

- We identify a trip-to-window label-granularity mismatch in driving behavior analysis. Using UAH-DriveSet, we show that windows assigned with inherited trip labels are poorly separated in feature space and that these labels primarily characterize entire trips rather than distinct behaviors within individual time windows.
- We propose DriveStateReg, a state recognition and semantic annotation framework that constructs time window-level behavioral states from ego-vehicle motion signals through unsupervised clustering, LLM-based annotation, and manual verification.
- We instantiate DriveStateReg on UAH-DriveSet and provide evidence that the recognized states provide clearer structure, stronger neighborhood consistency, richer within-trip behavioral composition, and interpretable sensor-grounded semantics compared with inherited trip labels.

2 Label Granularity Analysis

We start by addressing an important question: are trip-level labels appropriate for short segments? We analyze this issue using UAH-DriveSet [25], a widely used dataset in driving behavior analysis [9, 11, 19–21, 29, 30], which contains 40 trips collected from 6 drivers and provides three trip-level categories, namely Aggressive, Normal, and Drowsy. To avoid the influence of external factors, we only use ego-vehicle motion signals in the experiments. Each trip is segmented into 5s windows with a 1.25s stride, whose labels are inherited from their parent trips.

To quantify the relative dynamic level of driving behavior, we compute a dynamic score measuring the normalized feature deviation of each window. If trip-level labels could reliably serve as time window-level labels, windows with the same inherited label should have similar dynamic scores, whereas windows with different inherited labels should show clearly separated score distributions. However, Figure 1(a) reveals a clear label-granularity mismatch. The inherited Normal, Drowsy, and Aggressive window labels substantially overlap in the dynamic-score space, with an average overlap coefficient [15] of 0.733. This indicates that the three trip-level labels do not form clearly separated behavioral states.

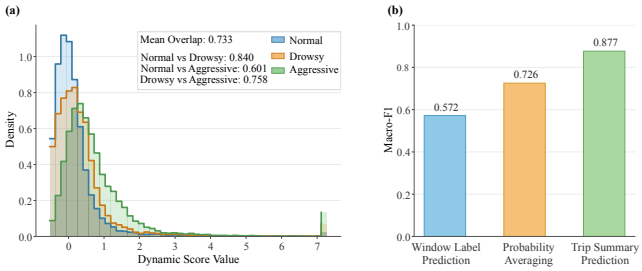


Figure 1: Diagnostics of trip-to-window label mismatch. (a) Dynamic-score overlap. (b) Macro-F1 comparison of three prediction pathways.

The detailed computation procedures are provided in our online repository.[1]

To further diagnose the label-granularity mismatch, we compare three prediction pathways in Figure 1(b): predicting inherited trip labels from individual window features, averaging time window class probabilities to obtain trip predictions, and predicting trip labels from statistical summaries of window features. The results show that direct window-level prediction achieves the lowest macro-F1, whereas both trip-level aggregation strategies perform better. This pattern suggests that the trip-level labels provide weak and potentially conflicting supervision for window-level driving behavior analysis, motivating the state recognition framework introduced next.

3 DriveStateReg: Driving State Recognition

To address the label-granularity mismatch between trip-level labels and time window-level driving behavior, we propose DriveStateReg, a state recognition framework that infers behavioral states from unlabeled ego-vehicle motion signals. The core motivation of DriveStateReg is to replace inherited trip labels by discovering time window behavioral states directly from sensor data, and then assigning interpretable semantic descriptions to these states using LLM-based annotation and manual verification. Figure 2 illustrates the overall framework, which consists of four steps.

a. Feature Extraction. To extract time window behavioral states from sensor data, the first step of DriveStateReg is to construct a feature space. Each trip is segmented into overlapping windows, and each window is represented using ego-vehicle motion signals such as speed, acceleration, heading change, attitude, and yaw. From these signals, we extract summary, variability, angular-change, and compound motion features, so that the representation captures both baseline motion patterns and short-term fluctuations in driving behavior. The resulting feature representation is standardized and reduced by PCA [17] before state recognition. In practice, applying our framework to UAH-DriveSet yields 62 features and a 22-dimensional PCA representation with 95% cumulative explained variance.

b. State Recognition. To address the absence of reliable time window-level labels, DriveStateReg formulates the annotation task as an unsupervised state recognition problem. We use *state system* to denote a complete time window-level partition produced by one

[1] https://github.com/zxy0303/DriveStateReg

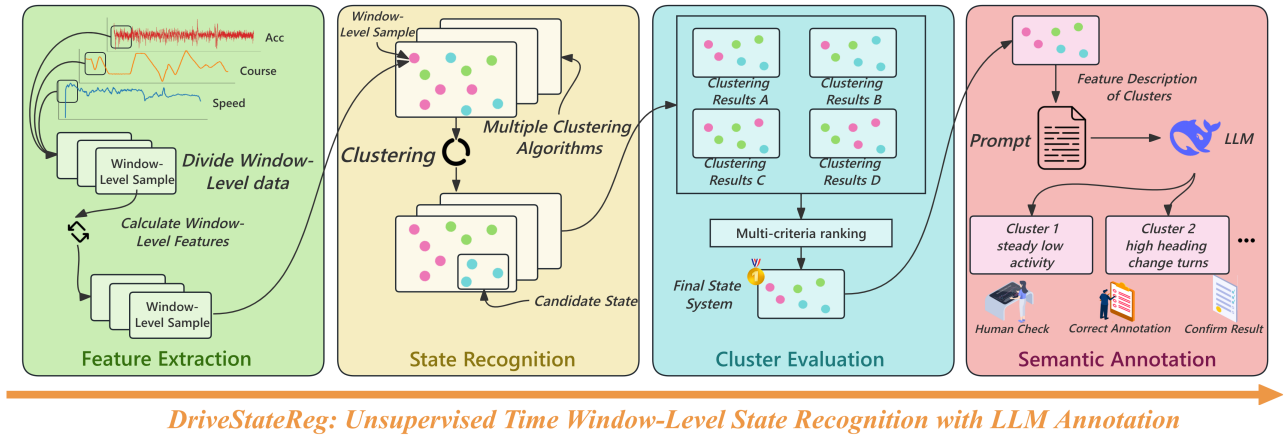


Figure 2: Time window-level state recognition and semantic annotation framework.

Table 1: Comparison between inherited trip labels (three states) and DriveStateReg's four states in the same feature space. MTE denotes median trip-internal entropy, and ESPT for effective states per trip, both measure within-trip behavioral heterogeneity.

State System	Silhouette	DB [6]	CH [2]	KNN purity@10	MTE	ESPT
Inherited trip labels (three states)	-0.01	9.94	130.70	0.54	0.00	1.00
DriveStateReg (four states)	**0.14**	**2.10**	**1262.61**	**0.89**	**0.77**	**2.16**

clustering configuration. Because the true number and geometry of time window behavioral states are unknown, DriveStateReg generates multiple candidate state systems under different unsupervised learning assumptions. We apply complementary unsupervised algorithms—including Agglomerative Clustering (various linkages) [23], Bisecting K-Means [31], divisive Gaussian Mixture Models (GMM) [7], and BIRCH [35]—varying the target cluster number $K \in \{2, ..., 8\}$.

c. Cluster Evaluation. To avoid choosing a state system only because it performs well under a single clustering index, DriveStateReg selects the final system through multi-criteria ranking. Accordingly, candidate systems are evaluated across multiple criteria: internal geometry (Silhouette [27], Davies-Bouldin [6], Calinski-Harabasz [2]); nearest-neighbor consistency via cluster purity [5, 22]; temporal persistence using self-transitions, run length [24], and shuffled null models [10]; intra-trip heterogeneity via state entropy and effective state count [13, 26]; cross-driver and cross-trip coverage; cluster stability under perturbation via Jaccard similarity [1, 12]; and distinctiveness from inherited trip labels using NMI [14, 32]. Each evaluation criterion produces a ranking over candidate state systems, and the rankings are aggregated using Borda-style rank aggregation [18]. The top-ranked candidate is selected for semantic annotation.

d. LLM-Based Semantic Annotation. To convert latent cluster indices into interpretable labels, DriveStateReg performs LLM-based semantic annotation followed by manual verification. For each selected state, we summarize structured evidence including state size and major feature deviations. These pieces of evidence are provided to the LLM through a constrained prompt, requiring the LLM to ground each proposed label in observable ego-vehicle motion patterns. We use DeepSeek-V4-Flash-Preview via the DeepSeek API (`deepseek-reasoner`) as the annotation model, which generates a preliminary state name, verification points, and confidence. The generated annotations are subsequently verified by the first author. For the manual-review set, representative segments are selected from trips with high state coverage, within-trip diversity, transition frequency, and inclusion of the least frequent state. Guided by the verification points generated by the LLM, the reviewer inspects the corresponding state timelines, feature trajectories, and raw sensor signals to assess whether the proposed semantic description is supported by observable motion patterns. When the observed evidence suggests a more comprehensive interpretation than the initial LLM-generated label, the state name is revised accordingly. The complete prompt and model configuration are provided in the online repository.

4 Evaluation of DriveStateReg

In the 22-dimensional representation space, DriveStateReg evaluates 238 candidate state systems. Under the multi-criteria protocol, a four-state partition generated by Bisecting K-Means is introduced, referred to as `DriveStateReg's four states`. Through LLM-based annotation and manual verification, the four states are named `Stable Cruising`, `Adaptive Roll-Longitudinal State`, `Active Lateral Adjustment`, and `Highly Dynamic Driving`.

We evaluate whether DriveStateReg provides a more suitable time window-level annotation than inherited trip labels. The evaluation is organized around three questions.

Q1. Do DriveStateReg's four states form more coherent time window-level driving state categories and better capture within-trip variation?

Table 1 compares DriveStateReg's four states with inherited trip labels in the same feature space. The results show that the state

Table 2: Scales, coverage, temporal persistence, duration, and manual verification support of `DriveStateReg's four states`. "Driver cov." denotes driver coverage; "Trip cov." denotes trip coverage; "Self-trans." denotes the self-transition probability between adjacent windows and "Mean dur." denotes the mean duration of consecutive windows assigned to the same state.

States	Quantity	Proportion	Driver cov.	Trip cov.	Self-trans.	Mean dur.	Manual Verification
Stable Cruising	5550	57.45%	6/6	40/40	0.87	7.39	5/5
Adaptive Roll-Longitudinal State	2194	22.71%	6/6	40/40	0.65	2.78	5/5
Active Lateral Adjustment	1523	15.77%	6/6	40/40	0.72	3.54	3/5
Highly Dynamic Driving	393	4.07%	6/6	38/40	0.68	3.09	4/5

system selected by our framework improves over inherited trip labels from several complementary perspectives. Silhouette, Davies-Bouldin (DB), Calinski-Harabasz (CH) and KNN purity indicate that DriveStateReg's four states produce a more organized time window-level feature space, with better cluster compactness, separation, and neighborhood consistency. Median trip-internal entropy and the effective number of states per trip quantify the within-trip variation represented by the discovered states, which inherited trip labels cannot express by construction. In addition, the NMI between DriveStateReg's four states and the original trip labels is only 0.04, in contrast to the value of 1.00 for inherited trip labels, suggesting that the discovered states are not a simple restatement of the trip-level categories. Therefore, the improvement is not limited to a single clustering score; rather, the criteria consistently indicate that DriveStateReg's four states provide a more coherent representation for time window-level driving analysis.

Q2. Do DriveStateReg's four states exhibit temporal stability within trips?

The temporal structure further indicates that the labels produced by DriveStateReg are not random window-wise noise produced by fragmentation. The self-transition probabilities and mean durations reported in Table 2 characterize the temporal persistence of each state: self-transition measures how often adjacent windows remain in the same state, while mean duration measures the average length of consecutive windows assigned to that state. The stable scale and broad coverage of each state further demonstrate that the four states are consistently observed across available drivers and trips, rather than being specific conditions of a particular driver or trip. Moreover, stability analyses also support the robustness of the selected system: the overall mean Jaccard scores under subsampling, feature-noise perturbation, and random-seed refitting are 0.82, 0.89, and 0.82, respectively. These results suggest that DriveStateReg's four states can form continuous segments with a certain degree of temporal persistence while still allowing state transitions within a trip.

Q3. Are the discovered states semantically interpretable and grounded in sensor evidence?

After determining DriveStateReg's four states, we assign semantic names through LLM-based annotation and manual verification. Following the manual verification procedure described in Section 3(d), one representative segment was inspected for each of the five selected trips. Figure 3 presents one such segment with DriveStateReg's four states over selected sensor-derived features. The verification score in Table 2 denotes the number of inspected segments whose sensor-derived patterns supported the final semantic label. These qualitative observations support the feature-based interpretation of the final state names.

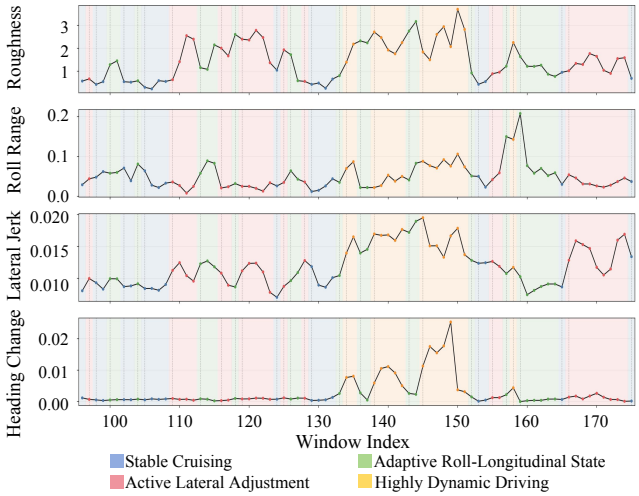


Figure 3: Representative timeline of DriveStateReg's four states over selected sensor-derived features.

5 Conclusion and Future Work

In this paper, we reveal how directly applying trip-level labels to short time windows obscures within-trip behavioral heterogeneity and yields weak, conflicting supervision for fine-grained driving analysis. To overcome this, we propose DriveStateReg, a framework that shifts annotation from top-down label inheritance to bottom-up state recognition. By integrating multi-criteria clustering evaluation with LLM-based semantic annotation, DriveStateReg can effectively extract fine-grained, interpretable behavioral states directly from unlabeled ego-vehicle motion data. Our evaluations on UAH-DriveSet demonstrate that these discovered states exhibit improved neighborhood consistency, temporal persistence, and sensor-grounded interpretability compared to inherited trip labels.

Our future work will focus on integrating DriveStateReg into practical in-vehicle intelligent systems. We will explore how the recognized driving states can be leveraged in downstream behavior-aware applications, as these discovered states offer a continuous characterization of behavioral evolution within a trip. This rich information can directly empower real-time monitoring, personalized feedback, and adaptive driver-assistance strategies to identify and respond to emerging behavioral changes. In addition, we plan to incorporate richer contextual information, such as road types and traffic conditions, and to evaluate and refine the framework using larger and more diverse datasets. These extensions may further improve state interpretability and robustness, while broadening the applicability of fine-grained driving state analysis in real-world driving scenarios.

References

[1] Asa Ben-Hur, Andre Elisseeff, and Isabelle Guyon. 2001. A stability based method for discovering structure in clustered data. In *Biocomputing 2002*. World Scientific, 6–17.

[2] Tadeusz Caliński and Jerzy Harabasz. 1974. A dendrite method for cluster analysis. *Communications in Statistics* 3, 1 (1974), 1–27.

[3] Dongyao Chen and Kang G. Shin. 2019. TurnsMap: Enhancing Driving Safety at Intersections with Mobile Crowdsensing and Deep Learning. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 3, Article 78 (2019), 22 pages.

[4] Yehoshua Shuki Cohen and Erez Shmueli. 2018. Money Drives: Can Monetary Incentives based on Real-Time Monitoring Improve Driving Behavior? *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 1, 4, Article 131 (Jan. 2018), 22 pages.

[5] Thomas M. Cover and Peter E. Hart. 1967. Nearest neighbor pattern classification. *IEEE Transactions on Information Theory* 13, 1 (1967), 21–27.

[6] David L. Davies and Donald W. Bouldin. 1979. A Cluster Separation Measure. *IEEE Transactions on Pattern Analysis and Machine Intelligence* PAMI-1, 2 (1979), 224–227.

[7] Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin. 1977. Maximum Likelihood from Incomplete Data via the EM Algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)* 39, 1 (1977), 1–22.

[8] Jair Ferreira Júnior, Eduardo Carvalho, Bruno V. Ferreira, Cleidson de Souza, Yoshihiko Suhara, Alex Pentland, and Gustavo Pessin. 2017. Driver behavior profiling: An investigation with different smartphone sensors and machine learning. *PLOS ONE* 12, 4 (2017), e0174959.

[9] Yan Gong, Jianli Lu, Wenzhuo Liu, Zhiwei Li, Xinmin Jiang, Xin Gao, and Xingang Wu. 2024. SIFDriveNet: Speed and Image Fusion for Driving Behavior Classification Network. *IEEE Transactions on Computational Social Systems* 11, 1 (2024), 1244–1259.

[10] Phillip I. Good. 2005. *Permutation, Parametric, and Bootstrap Tests of Hypotheses* (3rd ed.). Springer, New York, NY, USA.

[11] Bin Guo and John H. L. Hansen. 2024. T-EnFP: An Efficient Transformer Encoder-Based System for Driving Behavior Classification. In *ICASSP 2024 – 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 6830–6834.

[12] Christian Hennig. 2007. Cluster-wise assessment of cluster stability. *Computational Statistics & Data Analysis* 52, 1 (2007), 258–271.

[13] Mark O. Hill. 1973. Diversity and evenness: A unifying notation and its consequences. *Ecology* 54, 2 (1973), 427–432.

[14] Lawrence Hubert and Phipps Arabie. 1985. Comparing partitions. *Journal of Classification* 2, 1 (1985), 193–218.

[15] Henry F Inman and Edwin L Bradley Jr. 1989. The overlapping coefficient as a measure of agreement between probability distributions and point estimation of the overlap of two normal densities. *Communications in Statistics-Theory and Methods* 18, 10 (1989), 3851–3874.

[16] Landu Jiang, Xinye Lin, Xue Liu, Chongguang Bi, and Guoliang Xing. 2018. SafeDrive: Detecting Distracted Driving Behaviors Using Wrist-Worn Devices. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 1, 4, Article 144 (Jan. 2018), 22 pages.

[17] Ian T. Jolliffe and Jorge Cadima. 2016. Principal component analysis: a review and recent developments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 374, 2065 (2016), 20150202.

[18] Shili Lin. 2010. Rank aggregation methods. *Wiley Interdisciplinary Reviews: Computational Statistics* 2, 5 (2010), 555–570.

[19] Jing Liu, Yang Liu, Di Li, Hanqi Wang, Xiaohong Huang, and Liang Song. 2023. DSDCLA: Driving style detection via hybrid CNN-LSTM with multi-level attention fusion. *Applied Intelligence* 53, 16 (2023), 19237–19254.

[20] Wenzhuo Liu, Yan Gong, Guoying Zhang, Jianli Lu, Yunlai Zhou, and Junbin Liao. 2024. GLMDriveNet: Global–Local Multimodal Fusion Driving Behavior Classification Network. *Engineering Applications of Artificial Intelligence* 129 (2024), 107575.

[21] Wenzhuo Liu, Jianli Lu, Junbin Liao, Yicheng Qiao, Guoying Zhang, Jiayin Zhu, Bozhang Xu, and Zhiwei Li. 2024. FMDNet: Feature-Attention-Embedding-Based Multimodal-Fusion Driving-Behavior-Classification Network. *IEEE Transactions on Computational Social Systems* 11, 5 (2024), 6745–6758.

[22] Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. *Introduction to Information Retrieval*. Cambridge University Press, Cambridge, UK.

[23] Fionn Murtagh and Pedro Contreras. 2012. Algorithms for hierarchical clustering: an overview. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* 2, 1 (2012), 86–97.

[24] Lawrence R. Rabiner. 1989. A tutorial on hidden Markov models and selected applications in speech recognition. *Proc. IEEE* 77, 2 (1989), 257–286.

[25] Eduardo Romera, Luis M. Bergasa, and Roberto Arroyo. 2016. Need data for driver behaviour analysis? Presenting the public UAH-DriveSet. In *2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 387–392.

[26] Michael Roswell, Jonathan Dushoff, and Rachael Winfree. 2021. A conceptual guide to measuring species diversity. *Oikos* 130, 3 (2021), 321–338.

[27] Peter J. Rousseeuw. 1987. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *J. Comput. Appl. Math.* 20 (1987), 53–65.

[28] Fridulv Sagberg, Selpi, Giulio Francesco Bianchi Piccinini, and Johan Engström. 2015. A review of research on driving styles and road safety. *Human Factors* 57, 7 (2015), 1248–1275.

[29] Khaled Saleh, Mohammed Hossny, and Saeid Nahavandi. 2017. Driving behavior classification based on sensor data fusion using LSTM recurrent neural networks. In *2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 1–6.

[30] Xing Sheng, Jianrong Cao, Junzhe Zhang, Zhen Wang, and Zongtao Duan. 2025. Driving Behavior Classification Method Based on Fourier Transform Multimodal Fusion. *IEEE Transactions on Intelligent Transportation Systems* 26, 10 (2025), 15000–15012.

[31] Michael Steinbach, George Karypis, and Vipin Kumar. 2000. *A Comparison of Document Clustering Techniques*. Technical Report TR 00-034. Department of Computer Science and Engineering, University of Minnesota, Minneapolis, MN, USA.

[32] Alexander Strehl and Joydeep Ghosh. 2002. Cluster ensembles—a knowledge reuse framework for combining multiple partitions. *Journal of Machine Learning Research* 3 (Dec. 2002), 583–617.

[33] Mahan Tabatabaie, Suining He, and Xi Yang. 2022. Driver Maneuver Identification with Multi-Representation Learning and Meta Model Update Designs. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 2, Article 74 (2022), 23 pages.

[34] Johannes Tulusan, Thorsten Staake, and Elgar Fleisch. 2012. Providing Eco-Driving Feedback to Corporate Car Drivers: What Impact Does a Smartphone Application Have on Their Fuel Efficiency?. In *Proceedings of the 2012 ACM Conference on Ubiquitous Computing (UbiComp '12)*. Association for Computing Machinery, 212–215.

[35] Tian Zhang, Raghu Ramakrishnan, and Miron Livny. 1996. BIRCH: An efficient data clustering method for very large databases. In *Proceedings of the 1996 ACM SIGMOD International Conference on Management of Data (SIGMOD '96)*. Association for Computing Machinery, New York, NY, USA, 103–114.